
Integrable Systems — Exercise Sheet 3

The exterior derivative, Stokes, and the symplectic form

Variant: Questions only

Conventions throughout, as in Lecture 3: $\omega = \sum_i dq_i \wedge dp_i$ in that order; Hamilton's vector field solves $\iota_X \omega = dH$, with X in the *first* slot; on two dimensions $(a dx + b dy) \wedge (c dx + d dy) = (ad - bc) dx \wedge dy$; and $\iota_v(\mu \wedge \nu) = \mu(v)\nu - \nu(v)\mu$ for 1-forms μ, ν . On $\mathbb{R}^3$, $\text{vol} = dx \wedge dy \wedge dz$.

Exercise 1 (d in three dimensions). On $\mathbb{R}^3$ with coordinates (x, y, z) let

$$\alpha = yz dx + xz dy + xy dz, \quad \beta = -z dx + x^2 dy, \quad \gamma = x dy \wedge dz + y dz \wedge dx + z dx \wedge dy.$$

(i) Compute $d\alpha$. Is α exact? If it is, find a primitive. (ii) Compute $d\beta$, and then $d(d\beta)$ directly. Using the table in the lecture's reading on grad, curl and div, say which classical operator $d\beta$ computes, and check the answer against the classical formula. (iii) Compute $d\gamma$ and name the classical quantity its coefficient is. (iv) Compute $\alpha \wedge \beta$ and $d(\alpha \wedge \beta)$, and check the graded Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta - \alpha \wedge d\beta$. Would the check catch the rule written with +?

Exercise 2 (the circulation round a small square). Let $\alpha = P dx + Q dy$ with $P = xy^2$ and $Q = x^3$, and let $S_h = [a, a+h] \times [b, b+h]$, its boundary ∂S_h traversed anticlockwise. (i) Compute $\oint_{\partial S_h} \alpha$ exactly, edge by edge, as a polynomial in h . (ii) Show that $h^{-2} \oint_{\partial S_h} \alpha \rightarrow (\partial_x Q - \partial_y P)(a, b)$ as $h \rightarrow 0$, and that this limit is $d\alpha(\partial_x, \partial_y)$ at (a, b) : the lecture's reading of $d\alpha$ as circulation per unit area. (iii) Compute $\int_{S_h} d\alpha$ and compare it with (i). Which theorem have you verified? (iv) For a general smooth α , pair opposite edges and expand to first order to show $\oint_{\partial S_h} \alpha = h^2(\partial_x Q - \partial_y P)(a, b) + O(h^3)$, although each edge alone is of order h .

Exercise 3 (what the angle form sees). Let $\alpha = (x dy - y dx)/(x^2 + y^2)$ be the angle form on $\mathbb{R}^2 \setminus \{0\}$, and let $t \in [0, 2\pi]$ throughout. (i) Compute $\oint \alpha$ over $\gamma_a(t) = (3 + \cos t, \sin t)$, $\gamma_b(t) = (\cos 2t, \sin 2t)$, $\gamma_c(t) = (2 \cos t, -\sin t)$, and $\gamma_m(t) = (\cos mt, \sin mt)$ for an integer $m \geq 1$. For γ_a , find a primitive of α on the half-plane $x > 0$ instead of integrating. For γ_c you may use $\int_0^{2\pi} dt/(A^2 \cos^2 t + B^2 \sin^2 t) = 2\pi/(AB)$ for $A, B > 0$. (ii) Obtain the answer for γ_c again without integrating: write down a deformation from the unit circle traversed clockwise to γ_c , check that it never meets the origin, and apply the lecture's corollary on deformation invariance. (iii) The circle $r = 2$ traversed anticlockwise and the circle $r = 1$ traversed clockwise together make up the boundary of the annulus $A = \{1 \leq r \leq 2\}$, with its induced orientation. Compute both integrals and check Stokes' theorem on A . Why does the deformation corollary *not* say that the two integrals are equal? (iv) Now let γ be *any* smooth closed curve in $\mathbb{R}^2 \setminus \{0\}$, written in polar form $\gamma(t) = r(t)(\cos \theta(t), \sin \theta(t))$ with smooth $r > 0$ and θ on $[0, 2\pi]$. (Every such curve can be written so; take that as given.) Closedness forces $\theta(2\pi) - \theta(0) = 2\pi m$ for an integer m , the *winding number* of γ . Show that $\oint_\gamma \alpha = 2\pi m$, however the loop wanders or crosses itself, and that $F(t, s) = ((1-s)r(t)+s)(\cos \theta(t), \sin \theta(t))$, $s \in [0, 1]$, is a deformation from γ to a loop on the unit circle that never meets the origin.

Exercise 4 (pulling back a 2-form). (i) Let $F(u, v) = (x(u, v), y(u, v))$ be smooth. Show that $F^*(dx \wedge dy) = \det(\partial(x, y)/\partial(u, v)) du \wedge dv$, and deduce $dx \wedge dy = r dr \wedge d\theta$ in polar coordinates. (ii) Let $\nu > 0$, and on $\mathbb{R}^2$ minus the origin introduce (φ, I) , $I > 0$, by

$$q = \sqrt{2I/\nu} \sin \varphi, \quad p = \sqrt{2I\nu} \cos \varphi.$$

Show that $dq \wedge dp = d\varphi \wedge dI$, and that the choice $q = \sqrt{2I/\nu} \cos \varphi$, $p = \sqrt{2I\nu} \sin \varphi$ gives $-d\varphi \wedge dI$ instead. (iii) With the first choice and $H = \frac{1}{2}(p^2 + \nu^2 q^2)$, express H in (φ, I) and solve $\iota_X \omega = dH$ there, using $\omega = d\varphi \wedge dI$ and the lecture's sign test with φ in the place of q . Check at $\varphi = 0$ that increasing φ is the direction of the motion. (iv) Compute $\oint p dq$ round the curve $I = \text{const}$, traversed with φ increasing, and compare it with the area of the region the curve encloses.

Exercise 5 (*p dq on the pendulum's cylinder*). The pendulum $H = \frac{1}{2}p^2 - \cos q$ has phase space the cylinder $S^1 \times \mathbb{R}$, with q an angle defined modulo 2π . (i) q is not a function on the cylinder. Explain why dq and $p dq$ are nevertheless 1-forms there, and compute $d(p dq)$. Is $p dq$ closed on the cylinder? (ii) For $E > 1$ the level $H = E$ has a component γ_E on which $p > 0$: the pendulum rotates. Write $\oint_{\gamma_E} p dq$, traversed in the direction of the motion, as an integral over $q \in [0, 2\pi]$, and show that it lies strictly between $2\pi\sqrt{2(E-1)}$ and $2\pi\sqrt{2E}$. Deduce that the pullback of $p dq$ to the circle γ_E is closed but not exact. (iii) Now let $H = p$ on the same cylinder. Show that every level set $\{p = c\}$ is an invariant circle, and compute $\oint p dq$ round it in the direction of the motion. What does the level $c = 0$ say about the claim that the loop integral of $p dq$ round an invariant circle cannot vanish?

Exercise 6 (*odd dimensions are degenerate*). (i) Show that an antisymmetric $m \times m$ real matrix is singular when m is odd. Deduce that a manifold carrying a non-degenerate 2-form has even dimension. (ii) On $\mathbb{R}^3$ find every vector v with $\iota_v \beta = 0$, where $\beta = dx \wedge dy + dy \wedge dz + dz \wedge dx$. (iii) On $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) find every v with $\iota_v \beta' = 0$, where $\beta' = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. Is β' closed? Is it symplectic?

Exercise 7 ★. *A magnetic symplectic form*. On $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) and a real constant B , let

$$\omega_B = dq_1 \wedge dp_1 + dq_2 \wedge dp_2 + B dq_1 \wedge dq_2.$$

(i) Show that ω_B is symplectic. (ii) For $H = \frac{1}{2}(p_1^2 + p_2^2)$, solve $\iota_X \omega_B = dH$ for X , and solve the equations of motion it gives. Describe the orbits in the (q_1, q_2) -plane, and show that H is conserved — first by computing, then from the antisymmetry of ω_B alone. (iii) Let $B = b(q_1, q_2)$ be a smooth function instead. Is ω_B still symplectic? What if b depends on p_1 as well?